Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	31 January 2026
Team ID	[Enter Team ID]
Project Name	Global Fuel Prices and Taxation Analysis (2020 to 2026)
Maximum Marks	4 Marks

Functional Requirements:
Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)	Priority (High/Medium/Low)
1	Data Access & Integration	Connect to global_fuel_prices_Dataset.csv (27,468 records, 84 countries, 7 regions, 2020 to 2026) with fields for country, region, date, income level, petrol/diesel/LPG price, tax percentage and subsidy level.	High
2	Data Processing	Set date and numeric data types, assign Country as a geographic role, and score subsidy level (Low=1 to Very High=4). Check for missing values.	High
3	Business Analysis	Calculate average tax, average fuel price, subsidy score and price volatility (standard deviation / mean). Aggregate by country, region, year and income level. Generate a 12-month diesel price forecast.	High
4	Business Insights & Analytical Outputs	Provide 9 visuals: tax map, tax vs subsidy, subsidy vs price, diesel forecast, top 10 / bottom 10 countries, volatility heatmap, and tax, subsidy and price by income. Add 2 to 3 written insights per scenario.	High
5	User Interaction & Analysis	Year filter, click-to-filter between the map and tax vs subsidy chart, hover tooltips, and region and income comparison.	Medium
6	Data Update & Reporting	Refresh the data source to update all sheets. Export the dashboard as a packaged workbook (.twbx), image and PDF for reports.	Medium

Non-functional Requirements:
Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	**Usability**	Clear titles, readable labels, consistent colors and one row of charts per scenario so a non-technical user can understand each view.
NFR-2	**Security**	Uses public, aggregated country-level fuel data with no personal information. Shared files are read-only.
NFR-3	**Reliability**	Charts always reflect the connected dataset; calculated fields use fixed formulas so results are repeatable.
NFR-4	**Performance**	Dashboard loads and filters within a few seconds on a dataset of about 27,000 records.
NFR-5	**Availability**	Available any time as a packaged workbook or a published web version viewable in a browser.

FR No.	Non-Functional Requirement	Description
NFR-6	**Scalability**	New years, countries or fuel types can be added to the data source and the sheets update without rebuilding.